
Foundations of Generative AI & the AI Generalist Toolkit

Post-session notes, resources and prompt library

Day 1 · Session 1

Generative AI Mastermind · Outskill

- 🕒 **Session:** Day 1, Session 1 of the two-day Generative AI Mastermind
- 📁 **What's inside:** Every concept and tool from the session — plus ready-to-use prompt templates for each use case, so you can practice them yourself
- ✦ **How to use:** Skim once end-to-end, then pick one or two use cases and implement them this week

1. About This Document

This is your deep-dive companion to the first session of the Mastermind. The session was deliberately **breadth-first**: the goal was to show you the full landscape of what is possible with generative AI today, not to make you an expert in any single tool. These notes follow the same arc the session did — first *how* generative AI works, then *how to talk to it well*, then *which model to use where*, and finally a tour of the **AI Generalist toolkit**, one real use case at a time.

Every use case below includes a **ready-to-use prompt** you can copy as-is, then adapt to your own work — your industry, your role, your data. At the end you will find a **complete tool directory** with websites and use cases.

The mindset for these two days

You do not need to be technical to use generative AI — most of what was shown requires **zero code**. Don't try to master everything at once. If you leave this session writing better prompts and using one or two new tools, you have already moved ahead of most professionals.

2. What Is Generative AI & How Does It Work?

"You can drive a car without knowing how the engine works. But if you understand the engine, you become a better driver." That is why the session began with fundamentals: knowing what happens under the hood makes you a sharper, more deliberate user of every AI tool.

2.1 Where LLMs sit in the AI universe

Generative AI is a method of artificial intelligence that can create original content — text, images, video — by learning patterns from large datasets. It sits inside a nested hierarchy:

- **Artificial Intelligence** — the broadest idea of "intelligent machines" (the benchmark keeps shifting: first chess, then Go, now language).
- **Machine Learning** — systems that learn from data to make predictions, like an e-commerce app predicting your next purchase from past behaviour.
- **Deep Learning / Neural Networks** — brain-inspired networks that solve hard perception problems, like face recognition in photos.
- **Large Language Models (LLMs)** — models that understand, summarise and generate human-like language. These are the focus of the Mastermind.

2.2 What happens when you press Enter in ChatGPT

Try this yourself: open chatgpt.com and ask it to explain LLMs like you're five. The answer comes back as an analogy — a super-smart parrot that has read millions of books and become world-class at one game: *guess the next word*. Say "twinkle, twinkle, little..." and it says "star".

Practice prompt

✓ Prompt:

"Explain large language models to me like I am 5 years old."

Under the hood, five things happen in sequence:

- **1. Tokenization.** Your text is broken into tokens — the smallest units a model works with. Rule of thumb: **1 token ≈ 0.75 words** (so 1 million tokens ≈ 750,000 words). All AI pricing and limits are quoted in tokens, never words.
- **2. Embedding.** Tokens are converted into numbers (vectors), because computers only understand numbers. Words with similar meanings sit close together — "bank" and "money" are neighbours; "bank" and "concert" are far apart.
- **3. Self-attention.** Like the usher in an old cinema shining a torch on your row, attention tells the model which words in your input matter most, instead of treating every word equally.
- **4. Prediction.** The model predicts the most likely next word ("...a robo-friend who read every ____" → *book*, not *bank*), then builds the response word by word while staying coherent.
- **5. Variation.** Because it is a probabilistic mathematical model, asking the same question twice will give differently worded answers with broadly the same meaning. That is normal, not a bug.

3. Prompting & Context Engineering

Prompt engineering is writing a better instruction instead of a vague one. **Context engineering** goes further: *giving AI everything a smart human would need to do the job well*. A new employee can't ace their job on day one — not because they lack ability, but because they lack context. The same is true of AI.

3.1 The five layers of context — the "Magic Prompt"

Layer	What it answers
Identity context	Who is the AI acting as? (the persona or role)

Layer	What it answers
World context	What does the AI need to know about your situation, business and audience?
Task context	What exactly needs to happen?
Example context	What does a great output look like? (samples, references, tone)
Constraint context	What are the boundaries, rules and non-negotiables?

3.2 Worked example — a rent vs. buy article

Here is the framework applied layer by layer to a common writing task. Swap in your own audience, topic and tone:

Practice prompt (Magic Prompt structure)

✓ Prompt:

"Identity: You are a seasoned personal-finance editor writing on real estate matters and helping people make good home-related finance decisions.

World: The audience is readers of a personal-finance publication, aged majorly 22–60 years, but tailored towards middle-class working professionals who are planning to buy their first house.

Task: Write a 200-word essay on rent vs. buy a house in India in 2026.

Constraints: Keep it low on jargon. No fluff. A to-the-point, no-nonsense article cutting through the hype, with a dash of humour.

Example: You can follow the style of a personal-finance writer you admire."

The secret: never accept the first output

Always iterate with feedback. For example:

✓ Follow-up prompt:

"Try to add these additional pointers: customise it for people who earn ₹10–30 lakh per annum, warm pros and cons for rent versus buy, along with nuance on influencers who might be overhyping either renting or buying."

Vague vs. specific — why context wins

✗ "Write an article on renting vs buying a house." — generic, could be for anyone, anywhere.

✓ **Specific prompt:** *identity + audience + word count + tone constraints + a style reference* — produces a piece that sounds like it was written for your exact reader.

3.3 The AI skills ladder

Most people start as **basic users**. The journey the Mastermind takes you on: **Basic user → Prompter → Automations → Agents → Builders → AI Generalist**. Beyond the AI Generalist sit machine learning engineers (software) and researchers (math) — but the Generalist focuses purely on *applying* AI, and that is where the biggest career opportunity sits today.

4. Choosing the Right Model

The most commonly used assistants are **ChatGPT** (OpenAI), **Claude** (Anthropic) and **Gemini** (Google), with **Copilot** (Microsoft) common in business environments, **Meta AI**, and **Grok** (xAI) — which is genuinely underrated. But there are far more models in the world than most people guess.

4.1 OpenRouter — the model marketplace

On **openrouter.ai** you can see 400–600+ live models, their leaderboards, and — most usefully — **Categories by use case**. It doesn't tell you which model is "best"; it shows you which models real users actually route their finance, marketing, programming or legal prompts to. Check the category for *your* kind of work before choosing a model — the rankings shift continuously as models get upgraded.

4.2 GetMulti — compare models side by side

On **getmulti.ai** you add 2–4 models (for example Claude Opus, GPT, Gemini Pro and Grok) and send one prompt to all of them at once, watching responses stream side by side. Mark which response you prefer and over time you learn which model suits which of your tasks.

Practice comparison prompt

✓ Prompt:

"Explain what rising US 10-year Treasury yields mean for Indian and US stocks, especially IT, for a retail investor. Keep it under 200 words, no jargon, and end with a one-line hot take."

Any question from your own field works — the point is to watch how differently each model answers the same brief.

4.3 Reasoning ("thinking") models — the one type everyone must know

Reasoning models think before they answer — calibrating, searching, weighing trade-offs — and they are the most powerful class of model available. Examples: ChatGPT's *Thinking* mode, Claude Sonnet/Opus, Gemini Pro, DeepSeek R1, Grok Thinking.

When should you use a thinking model?

Simple calibration rule: **if you, as a human, would break your head over the task, let AI break its head too** — use a reasoning model. If it's a direct, simple task, an instant model is faster and perfectly fine. The more powerful the model, the longer it thinks — and the better the answer.

Practice reasoning prompt (use a thinking model)

✓ Prompt:

"I have a ₹40,000 surplus every month. Should I prepay my ₹35 lakh home loan at 8% interest with 10 years left, or invest the surplus in an equity index fund (assume 11% long-term return)? Consider tenure, interest rate, expected returns, risk, taxes and liquidity. Give me two contrasting strategies with the math, and a one-line verdict for a moderate-risk person."

A good reasoning model will compare end-states for both strategies, flag the real swing factors (tax regime, liquidity), and — importantly — **not make the decision for you**. It expands the calculation you would otherwise do in your head. The human stays in charge.

Bonus: turn the answer into a mini-app (Artifacts)

✓ Follow-up prompt:

"Build the interactive calculator." (then: "Make suitable assumptions and proceed.")

Claude will build a working **artifact** — a small interactive prepay-vs-invest calculator with sliders for loan outstanding, interest rate, tenure, surplus, expected equity return and tax regime, showing the verdict change live. You can download or publish artifacts and reuse them anytime.

4.4 Skills — turn your repeated process into an SOP

If you give AI the same sequence of prompts over and over, package the process once as a **skill** (a playbook the AI follows — like a chef's recipe that any chef can execute). For example, someone who regularly designs corporate training curricula can build a "curriculum" skill once, then run any new request through it in minutes and refine the output by hand. *AI takes the work to 70%; you bring it to 100%. What used to take 3–4 hours now takes about 45 minutes.*

Example skill invocation

✓ Prompt:

"/curriculum-builder — I want to build a curriculum for an enterprise client, for their marketing team's usage of AI. A quick 3-hour workshop."

Tasks like this run **in parallel** — you can keep working in other chats while the skill builds.

5. The AI Generalist Toolkit — Use Case by Use Case

An AI Generalist keeps a **stack**: a primary tool and a backup tool for each category of task (productivity, research, data, learning, marketing). When one tool has an outage, you switch. Below is every tool covered in the session, with a ready-to-use prompt. ⚠ Note: some of these (Wispr Flow, Fireflies, Phot.AI, Supergrow, Happenstance) are paid tools — the ROI argument is that incremental gains compound quickly.

5.1 Dictate instead of type — Wispr Flow

Typing slows your prompting down. **Wispr Flow** (Mac, Windows, iPhone) lets you dictate prompts anywhere — and unlike built-in dictation, it cleans grammar and formats your speech into polished text as you talk.

Practice prompt — dictate it instead of typing (ChatGPT, Thinking mode)

✓ Prompt:

"I have an investment thesis: I want to invest in mutual funds with exposure to the US NASDAQ 100 index, along with a mix of Indian portfolios that have exposure to semiconductor stocks. Can you do a roast of my investment thesis? Tell me where I am right, where I am wrong, what the risks are, and what I am completely missing."

✓ Follow-up:

"Too much information, not able to absorb. Can you give me the too-long-didn't-read version?"

Swap in any plan, thesis or idea of your own — asking AI to *roast* your thinking, then asking for the TL;DR, is a pattern that works everywhere.

5.2 Let AI attend your meetings — Fireflies.ai

Fireflies.ai auto-joins your Zoom / Google Meet calls, transcribes them, produces summaries and action items, and lets you interrogate any meeting you missed. It also connects to Claude via **Connectors** (Claude → Connectors → Browse → Fireflies), so you can query your meetings from inside your AI assistant.

Practice prompts

✓ Prompt 1:

"For this meeting — was everything covered properly?"

✓ Prompt 2:

"For this meeting, what aspects of [topic] were discussed?"

Use case: verify what was actually covered in a meeting you missed versus what you intended, without watching the recording.

5.3 Chat with YouTube videos & 400-page documents — NotebookLM

NotebookLM (notebooklm.google.com) answers *only* from the sources you give it — it does not consult the open internet, which means **no hallucinations** from outside material. Two powerful use cases:

- **YouTube:** paste any video URL as a source (NotebookLM and YouTube are both Google products) and chat with the video instead of watching 20 minutes.
- **Huge documents:** drop in a 400+ page document — say, the Draft Red Herring Prospectus of a recent IPO — and interrogate it for exactly what you need.

Practice prompts

✓ YouTube prompt:

"Give me a quick summary of all the key updates covered in the video."

✓ Large-document prompt:

"I want a quick summary of the financial situation and profitability of the company. Can you give me that summary?"

Every answer cites the exact paragraph of the source it came from — click the citation to verify.

5.4 Deep Research — ChatGPT & Claude

For anything where a single web search isn't enough — testing an investment thesis properly, or due diligence on a company you might join — use **Deep Research** (in ChatGPT) or **Research** (in Claude). Expect it to take **10–15 minutes** and run hundreds of searches; give it space to breathe and do other work in parallel. Every claim comes back with numbered citations.

Practice research prompts

✓ Investment thesis (Deep Research):

"Do a comprehensive deep research on my investment thesis: [describe your thesis]. Summarise the thesis, analyse the exposure, assess the relevant stock universe and mutual fund options, identify macro, sector and concentration risks, and draft a candid critique with strengths, weaknesses, missing elements and solutions."

✓ Job due diligence (Research):

"I am an AI generalist looking at a role at [Company]. Do a due diligence on the AI initiatives of [Company], plus a little bit on overall workplace feedback — so that I know there is enough real work and it's not just that they are hiring me with nothing to do. (Role context: [paste the job description or a short summary of the role].)"

A good research output includes a TL;DR, hiring signals, leadership narrative, employee-review sentiment, competitor comparison and a recommendation — each statement backed by a clickable source.

5.5 Practice tough conversations — Voice Mode mock interviews

Use the **voice mode** in ChatGPT, Claude or Gemini as a sparring partner for interviews, salary negotiations or hard client conversations. The flow: give full context in text first, then switch to voice and have the conversation out loud. Afterwards, ask for feedback — the whole mock interview is saved in the chat.

Practice prompts

✓ Setup prompt:

"I am preparing for an interview for a [role] at [Company]. The job involves [one or two lines on what the role does]. Assume a [senior interviewer title] is interviewing me for this role. Act as a completely tough interviewer and interview me. (Then switch on voice mode and begin.)"

✓ After the interview:

"Based on how it went — any feedback?"

✓ **Power move:** feed it your **LinkedIn profile summary** (see §5.10) before starting, so the interviewer grills you on your actual track record.

5.6 Data analysis without the paranoia — Claude dashboards

Two sides of the same fear: building dashboards, and reading them.

Practice prompt — build a dashboard from a CSV (Claude, Sonnet)

✓ Prompt:

"(Attach your dataset — e.g. sales data with transaction ID, date, product category, product name, units sold, unit price, total revenue, region, payment method.) Assume you are a McKinsey-level data analyst analysing this data. Go deeper into the data and build a small dashboard so that people who want a quick summary — especially my sales head — can absorb it at a glance."

Practice prompt — decode a scary dashboard from a screenshot

✓ Prompt:

"(Attach a screenshot of the dashboard.) For this dashboard, can you give me a quick summary of what it is telling me, and the insights and decisions I can take based on it?"

AI will explain what the dashboard measures, surface the key insights (e.g. where cash or attention is locked up), and translate them into concrete next decisions — even for dashboards you didn't build and don't fully understand.

5.7 Product photoshoots & ads — Phot.AI

Struggling with product photoshoots? **Phot.AI**'s AI Graphic Studio creates product ads from a product link or photo: it fetches the product, takes your feature description, pulls a brand's logo, colours and fonts automatically, and generates ad creatives and angles.

Example feature description (adapt to your product)

✓ Input:

"[Describe your product's standout features — e.g. for a smart kitchen appliance: 'A hybrid product which can predict gas usage; if there is an open flame it will auto-shut-off and prevent hazards. Elegant looking, easy to clean, sleek and durable.'](Then: 'Create an ad emphasising the [key benefit — e.g. safety] features.')"

⚠ Image generation takes a minute or two — expect some experimentation and the occasional failed first attempt.

5.8 Grow your LinkedIn in 5 minutes a day — Supergrow

Supergrow learns your writing style from your past posts and generates LinkedIn content that sounds like you — not like generic AI — then saves it as a draft for scheduling. Built specifically for the LinkedIn personal-branding use case: idea generation, posts, carousels, scheduling.

5.9 Find the person who can refer you — Happenstance

Happenstance connects to your networks (LinkedIn, Gmail, Twitter/X) and answers people-questions in natural language — searching not just your contacts but friends-of-friends to find warm paths.

Practice prompt

✓ Prompt:

"I am applying for a role at [Company]. Find people in my network who can refer me to this role at [Company]."

It will surface current employees, ex-employees and second-degree connections — exactly the people to message for a referral.

5.10 An AI-powered browser — Comet (Perplexity)

Comet, Perplexity's agentic browser, has a sidebar assistant that can *take control of the browser* — scrolling, clicking and reading on your behalf. A powerful pattern: have it crawl your own LinkedIn profile hands-free and produce a comprehensive professional summary, then paste that into ChatGPT voice mode for a sharper, context-rich mock interview. *That compounding — one tool's output feeding another — is the real AI Generalist skill.*

Practice prompts

✓ Prompt to the Comet assistant:

"Look at my complete LinkedIn profile, crawl through it, and give a comprehensive summary that I can give to AI tools to prepare for my interview."

✓ Then in ChatGPT:

"(Paste the summary.) Act as an interviewer for my application to [Company] — I'm interviewing for a [role]. Give me a tough interview based on what I've actually done."

5.11 Work with local files at scale — Claude Cowork

Every chat tool limits file uploads to a handful of files. When you have 25, 40 or 50 documents, use **Claude Cowork** — it opens a *local folder* on your machine as a project and reads files in parallel. A great first exercise: download the four quarterly shareholder letters of a listed company you've invested in (easily 200 pages combined) into one folder.

Practice prompt (Cowork, Sonnet, "act without asking")

✓ Prompt:

"I am an investor in [Company] — I hold a good amount of shares. In this folder are the four quarterly shareholder letters. Read the letters from Q1 to Q4, look at the growth of the business, and summarise the growth and the health of the business according to you."

Cowork reads all the documents in parallel and synthesises the full-year story: profitability trajectory, segment-by-segment health, cash position, and exactly what to watch next quarter.

5.12 Discover any tool for any job — There's An AI For That

"How do I keep up with all these tools?" — **theresanaiforthat.com** is a searchable directory of tens of thousands of AI tools by task. Searching a single word like "legal" surfaces hundreds of purpose-built tools instantly. Whenever a new use case appears in your work, search it here first.

6. Privacy, Accuracy & Keeping the Human in the Loop

The two most-asked audience questions, answered:

Privacy

If your data cannot leave your premises, use **open-source models running locally** (on your own GPUs/servers, no internet). There is maintenance work involved, but your data stays on-premise. (Running models locally is covered later in the Mastermind.)

Accuracy — especially for financial analysis

- **Prefer cited outputs.** Deep Research attaches a numbered source to every statement — open and verify them.
- **Do random manual checks.** Pick 2–3 statements from any AI report and verify them yourself. If your spot-checks pass, confidence in the whole rises. For data analysis, compute one answer manually and compare it with AI's.
- **Ask AI to audit itself.** Re-prompting *"Can you double-check your work and tell me whether you have done it correctly?"* catches and corrects a large share of errors.
- **Model Council (advanced).** Have multiple AI models review and debate each other's outputs.
- **Human in the centre.** AI expands the calculation; *you* make the decision. Never blindly act on AI financial output.

7. Tool Directory — Everything Covered in the Session

Tool	Use case	Website
ChatGPT	Everyday assistant; prompting, Thinking mode, Deep Research, Voice Mode mock interviews	chatgpt.com
Claude	Reasoning (Opus/Sonnet), Artifacts (mini-apps), Skills (SOPs), Projects, Connectors, data-analysis dashboards, Research	claude.ai
Gemini	Google's assistant; alternative for all the above use cases	gemini.google.com
Grok	xAI's assistant — crisp answers; genuinely underrated	grok.com
Microsoft Copilot	AI assistant widely used in business environments	copilot.microsoft.com

Tool	Use case	Website
Meta AI	Meta's consumer AI assistant	meta.ai
OpenRouter	Marketplace of 400–600+ models; leaderboards and rankings by use-case category (finance, marketing, coding...)	openrouter.ai
GetMulti	Send one prompt to multiple models and compare responses side by side	getmulti.ai
Wispr Flow	Voice dictation that auto-corrects grammar — dictate prompts instead of typing (Mac, Windows, iPhone)	wisprflow.ai
Fireflies.ai	AI meeting note-taker: auto-joins Zoom/Meet, transcripts, summaries, Q&A on meetings; connects to Claude	fireflies.ai
NotebookLM	Chat with YouTube videos and huge documents (e.g. 400-page IPO prospectus) — answers only from your sources, no hallucination	notebooklm.google.com
Phot.AI	AI product photoshoots and ad creatives from a product link/photo; auto-fetches brand logo, colours, fonts	phot.ai
Supergrow	Grow LinkedIn in 5 minutes a day — posts in your own writing style, carousels, scheduling	supergrow.ai
Happenstance	AI people search across your networks — find who can refer you to a job (incl. friends-of-friends)	happenstance.ai
There's An AI For That	Directory of AI tools by task — discover the right tool for any new use case	theresanaiforthat.com
Claude Cowork	Work on local folders with many files — read 25–50 documents in parallel (e.g. a year of shareholder letters)	claude.com/cowork
Comet (Perplexity)	Agentic AI browser — sidebar assistant that can control the browser, crawl pages and summarise on your behalf	perplexity.ai/comet

△ Wispr Flow, Fireflies, Phot.AI, Supergrow and Happenstance are paid tools (most with free tiers/trials). Model availability and rankings on OpenRouter change continuously — check current data before deciding.

8. Your Next Step

Closing advice from the session:

Don't be overwhelmed

Pick **one or two things** from this session and implement them this week. That is sufficient. Don't compare yourself with friends who know more AI — the only person to compare against is **who you were before the session**. If you have improved by 1%, congratulate yourself.

This session was a *taste*, not the full recipe — its job was to make you excited to learn the recipe. The sessions ahead go deeper: building your own AI tools and bots, agents, and running models locally.

See you in the next session. — Team Outskill